

Lecture - 7

* The direction of maximum rate of change for a fn. $J(\theta)$ is given by $\nabla_{\theta} J$.

$$\begin{aligned}\frac{\partial J}{\partial \vec{v}} &= \nabla J \cdot \vec{v} \quad [\vec{v} = \text{unit vector}] \\ &= |\nabla J| |\vec{v}| \cos \theta\end{aligned}$$

- For a given cost $|\nabla J|$ is fixed.

- $|\vec{v}|$ is also fixed for a direction.

To maximize $\frac{\partial J}{\partial \vec{v}}$, $\cos \theta$ needs to be 1.

$$\theta \rightarrow 0$$

$\vec{v}$ & ∇J are in parallel.

When you change a direction which is parallel to the gradient or along the gradient, you will increase most rapidly.

$$\theta_j := \theta_j - \alpha \frac{1}{n} \sum_{i=1}^n \left(\sum_{k=0}^M \theta_k x_k^{(i)} - y^{(i)} \right) x_j^{(i)}$$

$n = \text{no. of training examples.}$

Batch GD: - Consider all the training data to update the parameter.

- Slow when no. of training data is large.

Stochastic GD: - Considers one example at a time.

- We update the parameters according to the gradient of error wrt the single training example. (Incremental GD)

loop {

 for $i = 1$ to n {

$$\Rightarrow \theta_j := \theta_j + \alpha \left(y^{(i)} - \text{h}\theta(x^{(i)}) \right) x_j^{(i)} \quad (\text{for every } j)$$

}

}

- Stochastic GD is faster than Batch GD
- It may not converge to the minimum.

The Gradient :- $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$

$[A]_{m \times n}$

$$\nabla_A f(A) \in \mathbb{R}^{m \times n} = \begin{bmatrix} \frac{\partial f(A)}{\partial A_{11}} & \frac{\partial f(A)}{\partial A_{12}} & \dots & \frac{\partial f(A)}{\partial A_{1n}} \\ \frac{\partial f(A)}{\partial A_{21}} & \frac{\partial f(A)}{\partial A_{22}} & \dots & \frac{\partial f(A)}{\partial A_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f(A)}{\partial A_{m1}} & \frac{\partial f(A)}{\partial A_{m2}} & \dots & \frac{\partial f(A)}{\partial A_{mn}} \end{bmatrix}$$

$$(\nabla_A f(A))_{ij} = \frac{\partial f(A)}{\partial A_{ij}}$$

For a vector $\vec{x}$

$$\nabla_{\vec{x}} f(\vec{x}) = \begin{bmatrix} \frac{\partial f(\vec{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\vec{x})}{\partial x_n} \end{bmatrix}$$

$\nabla_{\vec{x}} [A\vec{x}] \rightarrow$ Not Defined

$$\rightarrow \nabla_{\vec{x}} (f(\vec{x}) + g(\vec{x})) = \nabla_{\vec{x}} f(\vec{x}) + \nabla_{\vec{x}} g(\vec{x})$$

$$\rightarrow f(\vec{x}) + t \in \mathbb{R} \quad \nabla_{\vec{x}} (t f(\vec{x})) = t \nabla_{\vec{x}} (f(\vec{x}))$$

The Hessian: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

$$H = \nabla_{\vec{x}}^2 f(\vec{x})$$

H is the $n \times n$ matrix of partial derivatives.

$$H = \nabla_{\vec{x}}^2 f(\vec{x}) \in \mathbb{R}^{n \times n} = \begin{bmatrix} \frac{\partial^2 f(\vec{x})}{\partial x_1^2} & \frac{\partial^2 f(\vec{x})}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f(\vec{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(\vec{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\vec{x})}{\partial x_2^2} & \dots & \frac{\partial^2 f(\vec{x})}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\vec{x})}{\partial x_n \partial x_1} & \frac{\partial^2 f(\vec{x})}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 f(\vec{x})}{\partial x_n^2} \end{bmatrix}$$

$$(\nabla_{\vec{x}}^2 f(\vec{x}))_{ij} = \frac{\partial^2 f(\vec{x})}{\partial x_i \partial x_j}$$

Hessian is always symmetric. $\frac{\partial^2 f(\vec{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\vec{x})}{\partial x_j \partial x_i}$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$\frac{\partial^2 f(x)}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial f(x)}{\partial x}$$

$\Rightarrow$ For a fn of a vector

$$\nabla_n \nabla_n f(x) = \nabla_n \left[\frac{\partial f}{\partial x_1} \frac{\partial f}{\partial x_2} \dots \frac{\partial f}{\partial x_n} \right]^T$$

Not defined

Hessian $\neq$ Gradient of gradients.

$$(\nabla_n f(x))_i = \frac{\partial f}{\partial x_i}$$

$$\nabla_n \frac{\partial f}{\partial x_i} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_i \partial x_1} \\ \frac{\partial^2 f}{\partial x_i \partial x_2} \\ \vdots \\ \frac{\partial^2 f}{\partial x_i \partial x_n} \end{bmatrix}$$

$\leftarrow$ i-th column of the Hessian.

$$\nabla_n^2 f(x) = \left[\nabla_n (\nabla_n f(x))_1, \nabla_n (\nabla_n f(x))_2, \dots, \nabla_n (\nabla_n f(x))_n \right]$$

$$\nabla_n^2 f(x) = \nabla_n (\nabla_n f(x))^T$$